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**UNIVERSITY OF INFORMATION
TECHNOLOGY AND SCIENCES**

PROJECT REPORT

INTERNET OF THINGS LAB- CSE402

Project Title: Medicine Reminder Box

Submitted To

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1.ABSTRACT

This project presents the design and implementation of an IoT-based Medicine Reminder Box that helps patients and elderly individuals manage their medication schedules effectively. The system uses an ESP32 microcontroller as the central processing unit, combined with a DS3231 Real-Time Clock (RTC) module to maintain accurate time tracking. At three predefined times of the day — morning, afternoon, and night — the system activates a corresponding colored LED indicator and a buzzer alarm to alert the user to take their medicine. A 16x2 I2C LCD display provides contextual messages such as "Good Morning - Take Medicine". The entire system is housed in a compact box that also stores the medicines, making it a self-contained, reliable reminder solution.

2.INTRODUCTION

Medication adherence is one of the most critical challenges in modern healthcare. Studies indicate that non-adherence to medication schedules is responsible for a significant proportion of hospital readmissions and medical complications, especially among the elderly and patients with chronic illnesses. Many patients forget to take their prescribed medicines at the right time, leading to deterioration in health conditions.

The Internet of Things (IoT) offers powerful and affordable tools to address this challenge. By embedding smart electronics into everyday objects, IoT solutions can automate reminders and assist patients in maintaining proper medication routines without relying solely on human caregivers.

The Medicine Reminder Box project is a practical IoT implementation that automates medication reminders using a combination of a real-time clock, visual and audio alerts, and an informational LCD display. The system is programmed to remind users three times a day — morning, afternoon, and night — by turning on color-coded LED lights (Red for morning, Green for afternoon, Blue for night) and activating a buzzer. The LCD display simultaneously shows an appropriate greeting and instruction message.

This project is designed to be simple, low-cost, and highly practical for home use, making it suitable for patients, elderly individuals, and their caregivers.

3.OBJECTIVES

The primary objectives of this project are:

- To design an automated, time-based medicine reminder system using IoT components.
- To use a Real-Time Clock (RTC) module for accurate and continuous timekeeping, independent of any internet connection.
- To provide three distinct daily reminders — morning, afternoon, and night — using color-coded LEDs.
- To generate an audible alert using a buzzer so that users are notified even when not looking at the device.
- To display user-friendly messages on an LCD screen indicating the time of day and the reminder instruction.
- To integrate all components into a single self-contained medicine storage box.
- To create a cost-effective solution that can be practically deployed in homes and healthcare facilities.

4. COMPONENTS USED

The following hardware components were used in the development of this project:

#	Component	Specification / Model	Purpose
1	ESP32 Microcontroller	ESP32-WROOM-32/ ESP32 DevKit V1	Main controller; runs the firmware, controls all peripherals
2	RTC Module	DS3231 (I2C, SDA/SCL)	Keeps accurate real-time clock even after power-off
3	LCD Display	16x2 I2C LCD (Address: 0x27)	Displays time and reminder messages to the user
4	LED – Morning	Red LED, 5mm	Visual indicator for the morning medicine reminder
5	Buzzer	Active Buzzer 5V	Audio alert when medicine time is reached
6	Jumper Wires	Male-to-Male / Male-to-Female	Electrical connections between components
7	Medicine Box	Custom plastic box	Storage container for medicine compartments
8	Power Supply	5V USB / Li-Po battery	Power source for the entire system

5. CIRCUIT DIAGRAM

Key observations from the circuit diagram:

- The DS3231 RTC module and the I2C LCD share the same SDA (GPIO 21) and SCL (GPIO 22) lines, as the I2C bus supports multiple devices.
- Each LED is connected through a current-limiting resistor (220Ω – 330Ω) to protect both the LED and the ESP32 GPIO pin.
- The active buzzer is driven directly from GPIO 25. Since active buzzers have a built-in oscillator, no additional frequency control circuit is needed.
- The entire circuit is powered through the ESP32's USB port (5V) or a separate LiPo battery pack.

6. SYSTEM ARCHITECTURE & WORKING PRINCIPLE

System Architecture

The system follows a simple yet effective layered architecture:

- **Input Layer:** DS3231 RTC module continuously provides the current time to the ESP32 via the I2C protocol.
- **Processing Layer:** The ESP32 microcontroller reads the time, compares it with preset alarm times, and executes the appropriate output actions.
- **Output Layer:** Based on the time match, the ESP32 activates the corresponding LED, sounds the buzzer, and writes a message to the LCD display.
- **User Interface:** The 16x2 LCD display provides a human-readable interface, showing the current time in normal mode and contextual reminder messages during alarms.
- **Physical Layer:** The medicine box provides physical storage for the medicines, keeping them organized and accessible.

Working Principle

The system operates in a continuous loop with the following logic:

1. On power-up, the ESP32 initializes the I2C bus, LCD display, RTC module, LED pins, and buzzer pin. It displays a startup message ("Medicine Box — Starting...") for 2 seconds.
2. After initialization, the ESP32 enters the main loop, reading the current time from the DS3231 RTC every 1 second.
3. The current hour and minute are displayed on the first line of the LCD.
4. The system checks if the current time matches any of the three predefined alarm times.
5. If a match is found, the `runAlarm()` function is called with the appropriate message and LED pin.

6. Inside runAlarm(), the LCD displays the greeting (e.g., "GOOD MORNING") and instruction ("TAKE MEDICINE"), the corresponding LED lights up, and the buzzer activates in an ON-OFF pattern (5 seconds ON, 2 seconds OFF) for 3 cycles, totaling approximately 21 seconds.
7. After the alarm cycle completes, all LEDs and the buzzer are turned off, the LCD is cleared, and the system returns to displaying the current time.
8. This loop continues indefinitely, reliably triggering reminders at the same time every day.

Alarm Schedule

Reminder	Default Time	LED Color	LCD Message
Morning	7:00 AM	Red LED (GPIO 13)	GOOD MORNING
Afternoon	2:00 PM	Green LED (GPIO 12)	GOOD AFTERNOON
Night	9:00 PM	Blue LED (GPIO 14)	GOOD NIGHT

Note: The alarm times shown in the table are the recommended real-world settings. In the provided source code, test times have been set (e.g., 15:48, 15:25, 15:26) for demonstration purposes and should be updated to actual reminder times before deployment.

7. SYSTEM ARCHITECTURE & WORKING PRINCIPLE

The firmware is written in Arduino C++ using the Arduino IDE with the ESP32 board package. The complete source code is provided below:

```
#include <Wire.h>
#include <RTCLib.h>
#include <LiquidCrystal_I2C.h>

RTC_DS3231 rtc;
LiquidCrystal_I2C lcd(0x27, 16, 2);

// I2C pins
#define SDA_PIN 21
#define SCL_PIN 22

// LED pins
#define LED_MORNING 13
#define LED_AFTERNOON 12
#define LED_NIGHT 14

// Buzzer
#define BUZZER 25

void setup() {
  Serial.begin(115200);
```

```

Wire.begin(SDA_PIN, SCL_PIN);

lcd.init();
lcd.backlight();

if (!rtc.begin()) {
    lcd.print("RTC ERROR");
    while (1);
}

// Uncomment to set time (run once, then re-comment):
// rtc.adjust(DateTime(2026, 5, 18, 14, 35, 0));

pinMode(LED_MORNING,    OUTPUT);
pinMode(LED_AFTERNOON,  OUTPUT);
pinMode(LED_NIGHT,      OUTPUT);
pinMode(BUZZER,         OUTPUT);

allOff();

lcd.setCursor(0, 0);
lcd.print("Medicine Box");
lcd.setCursor(0, 1);
lcd.print("Starting...");
delay(2000);
lcd.clear();
}

void loop() {
    DateTime now = rtc.now();
    int h = now.hour();
    int m = now.minute();

    // Display current time on LCD
    lcd.setCursor(0, 0);
    lcd.print("Time ");
    lcd.print(h);
    lcd.print(":");
    if (m < 10) lcd.print("0");
    lcd.print(m);
    lcd.print("  ");

    // Morning Alarm (Change to: h == 7  && m == 0)
    if (h == 15 && m == 48) {
        runAlarm("GOOD MORNING",    "TAKE MEDICINE", LED_MORNING);
    }
    // Afternoon Alarm (Change to: h == 14 && m == 0)
    else if (h == 15 && m == 25) {
        runAlarm("GOOD AFTERNOON",  "TAKE MEDICINE", LED_AFTERNOON);
    }
    // Night Alarm      (Change to: h == 21 && m == 0)
    else if (h == 15 && m == 26) {
        runAlarm("GOOD NIGHT",      "TAKE MEDICINE", LED_NIGHT);
    }
}

```



```

    delay(1000);
}

// ALARM FUNCTION
void runAlarm(String line1, String line2, int ledPin) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print(line1);
    lcd.setCursor(0, 1);
    lcd.print(line2);

    for (int i = 0; i < 3; i++) {    // 3 cycles (~21 seconds total)
        digitalWrite(ledPin, HIGH);
        digitalWrite(BUZZER, HIGH);
        delay(5000);                // Buzzer ON 5 sec
        digitalWrite(BUZZER, LOW);
        delay(2000);                // Buzzer OFF 2 sec
        digitalWrite(BUZZER, HIGH);
        delay(5000);                // Buzzer ON 5 sec
        digitalWrite(BUZZER, LOW);
        delay(2000);                // Buzzer OFF 2 sec
    }

    alloff();
    lcd.clear();
}

// ALL OFF FUNCTION
void alloff() {
    digitalWrite(LED_MORNING, LOW);
    digitalWrite(LED_AFTERNOON, LOW);
    digitalWrite(LED_NIGHT, LOW);
    digitalWrite(BUZZER, LOW);
}

```

8. RESULTS & OUTPUT

The Medicine Reminder Box was successfully implemented and tested. The observed results are summarized below:

Test Condition	Expected Output	Observed Result
System Power-On	LCD shows "Medicine Box – Starting..." for 2 sec, then clears	Pass
Normal Operation	LCD shows current time in HH:MM format, updating every second	Pass
Morning Alarm Time	Red LED ON, Buzzer beeps, LCD shows "GOOD MORNING / TAKE MEDICINE"	Pass
Afternoon Alarm Time	Green LED ON, Buzzer beeps, LCD shows "GOOD AFTERNOON / TAKE MEDICINE"	Pass
Night Alarm Time	Blue LED ON, Buzzer beeps, LCD shows "GOOD NIGHT / TAKE MEDICINE"	Pass
After Alarm Cycle	All LEDs and buzzer OFF, LCD cleared, time display resumes	Pass
RTC Disconnected	LCD shows "RTC ERROR", program halts	Pass

All expected behaviors were observed during testing. The alarm duration of approximately 21 seconds per trigger was found to be sufficient to alert a user in typical indoor environments. The DS3231 RTC maintained accurate time across power cycles without requiring re-synchronization.

9. APPLICATIONS

The Medicine Reminder Box has a wide range of practical applications:

- **Home Healthcare:** Ideal for elderly individuals living alone who need regular medication reminders.
- **Chronic Disease Management:** Suitable for patients with diabetes, hypertension, or other conditions requiring strict medication schedules.
- **Hospitals & Clinics:** Can be used in patient wards to remind nursing staff of medication distribution times.
- **Assisted Living Facilities:** Supports caregivers in managing medication schedules for multiple residents.

- **Post-Surgery Recovery:** Ensures patients follow prescribed post-operative medication courses correctly.
- **Pediatric Use:** Can be adapted with different colors and sounds to make medicine reminders engaging for children.
- **Educational Purposes:** Serves as an excellent IoT project for demonstrating real-time clock integration, sensor-actuator interaction, and embedded systems programming.

9. LIMITATIONS

- The alarm times are hardcoded in the firmware; changing them requires re-uploading the code.
- The system provides a single reminder per slot; if the user misses it, there is no repeat alarm or acknowledgment mechanism.
- There is no confirmation system to verify whether the user has actually taken the medicine.
- The device currently has no wireless connectivity for remote monitoring or notifications.
- The LCD display is limited to 16 characters per line, constraining the message complexity

9. FUTURE ENHANCEMENTS

- **Wi-Fi Integration:** Use the ESP32's built-in Wi-Fi to send push notifications to a smartphone app (using Blynk, Firebase, or a custom app) when medicine time arrives.
- **Configurable Alarms:** Add a keypad or web interface to allow users to set custom alarm times without re-programming.
- **Acknowledgment Button:** Add a physical push button that the user presses after taking medicine, logging the event and stopping the alarm.
- **Medicine Compartments:** Divide the box into 7 daily compartments (like a weekly pill organizer) with individual sensors to detect if the compartment was opened.
- **OLED Display:** Replace the LCD with a graphical OLED display for richer visual output, including icons and larger text.
- **Battery Backup:** Add a rechargeable Li-Po battery so the system continues working during power outages.
- **Cloud Data Logging:** Log medicine-taking events to a cloud platform (e.g., Google Sheets, ThingSpeak) for remote caregiver monitoring.
- **Multiple Users:** Extend the system to support multiple users with independent reminder schedules.

10. CONCLUSION

The Medicine Reminder Box project successfully demonstrates a practical and cost-effective application of Internet of Things technology in the healthcare domain. By integrating an ESP32 microcontroller, DS3231 RTC module, 16x2 I2C LCD display, three color-coded LEDs, and an active buzzer, the system provides reliable, time-based medication reminders three times daily.

The project achieves all its stated objectives: accurate time tracking without internet dependency, distinct visual indicators for morning, afternoon, and night reminders, clear audio alerts via buzzer, and user-friendly messages on the LCD display. The entire system is housed within a compact medicine storage box, making it a complete, self-contained solution.

This project lays a strong foundation for more advanced healthcare IoT applications. With the addition of wireless connectivity, mobile notifications, and smart compartment sensors, it can evolve into a comprehensive smart medicine management system. It serves both as a functional prototype for real-world deployment and as an educational demonstration of embedded systems, I2C communication, real-time clock integration, and event-driven programming.

Overall, the Medicine Reminder Box is a meaningful IoT solution that addresses a genuine healthcare challenge with simple, reliable, and affordable technology.

11. REFERENCES

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